M.S. Cybersecurity Candidate | SFS CyberAICorps scholarship recipient | UAB

Andrew Strachan

M.S. Cybersecurity student & builder

Dedicated to protecting national security and critical digital infrastructure. Bridging zero-trust engineering, on-device artificial intelligence, and pharmaceutical cGMP regulatory compliance with award-winning technical education leadership.

Interactive Resume & Technical Profile

Candidate credentials evaluated through interactive technology. Filter competencies across technical domains, review chronological stages, and inspect verified STAR accomplishment evidence.

Andrew Strachan

SFS CyberAICorps scholarship recipient (NSF-funded, with a post-graduation government service commitment). Preparing for Information Security Analyst roles.

Location: Birmingham, Alabama · Contact: astrachan162@gmail.com · Willing to obtain a security clearance.

Academic Record & Degree Verification

Master of Science

University of Alabama at Birmingham — M.S. Cybersecurity

M.S. in Cybersecurity · GPA: 3.75 / 4.0 · SFS CyberAICorps scholarship recipient (NSF-funded, with a post-graduation government service commitment)

Dates: January 2026 – Present (Expected Graduation: Dec 2027).

Key coursework: CS 623 Network Security, CS 646 Blockchain & Cryptocurrency, CS 636 Computer Security, CS 532 Systems Programming, and Cloud Computing (Summer 2026).

> **Career Impact:**
> Direct alignment with federal cyber defense architectures and zero-trust engineering standards.

State certificate

Provisional Certificate in a Teaching Field: Business, Marketing, and Finance (non-degree)

Provisional Certificate in a Teaching Field: Business, Marketing, and Finance (non-degree) · GPA: 3.66 / 4.0

Dates: Fall 2024 – Spring 2025 · University of Montevallo.

Focus: Managerial finance, enterprise risk frameworks, curriculum engineering, and Alabama TAPE standards.

> **Career Impact:**
> Educational governance, public-sector financial modeling, and structured communication rigor.

Bachelor of Science

B.S., Biochemistry (ACS), with a secondary major in Biology Medical Sciences; Minor in Music, Mississippi College (now Mississippi Christian University), Aug 2011 – May 2016, cum laude

B.S., Biochemistry (ACS), with a secondary major in Biology Medical Sciences; Minor in Music, Mississippi College (now Mississippi Christian University), Aug 2011 – May 2016, cum laude · GPA: 3.50 / 4.0

Dates: Aug 2011 – May 2016 · American Chemical Society Certified.

Activities: Senior Class Senator, Symphonic Band Principal Horn, Phi Mu Alpha Sinfonia co-founder, Delta Epsilon Iota Academic Honor Society, pre-medical rigor.

> **Career Impact:**
> Analytical methodology and problem decomposition.

Medical Studies, Course and clinical work

Medical Studies, Course and clinical work

Medical Studies, Course and clinical work, University of Mississippi School of Medicine, 2016–2020

Dates: 2016–2020

Activities: Human Anatomy Prosector (May–Sep 2017), President & Founder: P.A.L.S. (Peer-Assisted Learning Society); President: Quality Improvement Student Interest Group; Chair: UMMC/MBN Opioid Crisis Council; Virginia Covington Award.

> **Career Impact:**
> High-reliability clinical operations, crisis mitigation, bio-risk evaluation, and systemic triage.

Filterable Technical Competency Matrix

▶ STRIDE Threat Modeling	ZERO-TRUST	▶ Zero-Trust Network Architecture	ZERO-TRUST
▶ Wireshark & Packet Inspection	ZERO-TRUST	▶ Nmap Port & Vulnerability Scanning	ZERO-TRUST
▶ SHA-256 Block Hashing Primitives	ZERO-TRUST	▶ NIST SP 800-53 / CSF 2.0 Compliance	ZERO-TRUST
▶ CS 532 Systems Programming	ZERO-TRUST	▶ On-Device Local RAG Architecture	AI / ML
▶ Apple MLX & Metal Performance Shaders	AI / ML	▶ Quantized Gemma 4 / Ollama Execution	AI / ML
▶ SwiftData & Local SQLite Vector Search	AI / ML	▶ Model Context Protocol (MCP) Sandboxing	AI / ML
▶ Cloud Computing (Summer 2026)	CLOUD	▶ AWS S3, CloudFront, Lambda, DynamoDB	CLOUD
▶ Firebase Hosting, Firestore, Functions	CLOUD	▶ Expiring HMAC Token Verification	CLOUD

Chronological Professional Work History

Python Coding Summer Camp Teaching Assistant, UAB

Jun 2026 – Jul 2026

University of Alabama at Birmingham (Department of Computer Science)

- Instructed middle school students in fundamental Python programming concepts, data structures, and algorithmic logic.
- Guided students to program BBC micro:bit microcontrollers to interface with sensors, LED matrices, and external robotics.
- Mentored students in writing custom Python flight trajectory scripts for aerial drones, demonstrating physical computing and hardware-software integration.

CTE Teacher — Business, Marketing, and Finance

July 2024 – June 2025 (Full-Time, 40 hrs/wk)

Shades Valley High School (Jefferson County Board of Education - JEFCOED)

- Instructed Business Finance, Customer Service, and Sales in the school's premier National Academy of Finance (NAOF) program.

Career Research & Future Outlook

Labor market data for the Information Security Analyst role he is preparing for.

Target career he is preparing for: Information Security Analyst

Classification: **BLS SOC 15-1212.00** **CyberCorps SFS Public Service Specialization**

Information Security Analysts plan, engineer, implement, and monitor defensive security measures to protect computer networks, cloud architectures, and digital assets. Aligned with the National Initiative for Cybersecurity Education (NICE) Workforce Framework categories: **NICE PR-CD-001 (Cyber Defense Analyst)** **NICE SP-ARC-002 (Security Architect)** .

Zero-Trust Network Access **STRIDE Threat Modeling** **NIST SP 800-53 Controls**

Vulnerability Remediation **Post-Quantum Cryptography**

Official BLS Economic Research & Job Outlook (SOC 15-1212.00)

$120,360

MEDIAN ANNUAL WAGE

$57.87 / hour (BLS OOH National Median)

32%

PROJECTED GROWTH (2022–32)

Much faster than average (all jobs: 3%)

$182,370+

TOP DECILE WAGE (90TH %)

$87.68+ / hour (top decile, BLS)

+53,200

PROJECTED NEW JOBS

~16,800 average annual job openings

Industry (BLS SOC 15-1212.00) Federal (GS pay bands)

EARNINGS PERCENTILE	ANNUAL WAGE (USD)	HOURLY EQUIVALENT	PROFESSIONAL CONTEXT & TITLE
Lowest 10%	**$69,310**	$33.32 / hr	Junior Security Analyst / Associate SOC Analyst
25th Percentile	**$89,040**	$42.81 / hr	Cyber Defense Analyst / Network Security Specialist
50th Percentile (Median)	**$120,360**	$57.87 / hr	Information Security Analyst / Incident Responder
75th Percentile	**$152,430**	$73.28 / hr	Senior security analyst / threat modeler
Highest 10% (Top Decile)	**$182,370+**	$87.68+ / hr	Principal Security Architect / Federal Technical Lead

Industry Obstacles & Engineering Mitigations

CRITICAL INDUSTRY CHALLENGE 01

Post-Quantum Cryptography Migration

Quantum computer scalability threatens RSA and ECC asymmetric encryption, enabling 'harvest now, decrypt later' adversary tactics against federal systems.

> **Mitigation:**
> Research into NIST-standardized lattice-based algorithms (ML-KEM, ML-DSA). Engineered in CS 646 Sanctum 3D cryptographic visualization engine.

CRITICAL INDUSTRY CHALLENGE 02

Agentic AI Weaponization & Automated Exploits

Adversaries deploy autonomous LLM agents that discover zero-day bugs, synthesize polymorphic malware, and evade traditional firewall rules.

> **Mitigation:**
> Architected 'Citadel' zero-trust gatekeeper (Java 21 / MCP) enforcing sandboxed tool authorization and privacy-first local models (MaQkrs AI Tutor).

CRITICAL INDUSTRY CHALLENGE 03

Critical Infrastructure Ransomware Defense

National SCADA, healthcare, and educational IT systems face sophisticated ransomware cartels targeting unsegmented legacy networks.

> **Mitigation:**
> Enforcing defense-in-depth, STRIDE threat models, and pharmaceutical cGMP-derived audit SOPs to support operational continuity.

Career-Related Education

Academic coursework, research projects, and business education applied to real systems, accompanied by detailed analysis of their direct impact on Andrew's future cybersecurity career.

UAB Cybersecurity Graduate Coursework

CS 623 Network Security, CS 646 Blockchain & Cryptocurrency, CS 636 Computer Security, CS 532 Systems Programming, and Cloud Computing (Summer 2026).

Graduate coursework in those UAB courses.

> **Impact on Future Career:**
> Directly establishes the technical defense capabilities required for Information Security Analysts. Imparts hands-on mastery in packet inspection, TLS handshake negotiation, cryptographic primitives, and zero-trust cloud architectures.

CS 636 Computer Security

Enterprise Systems Security Architecture & Attack Boundary Deconstruction

Authored structured Level 0 and Level 1 Data Flow Diagrams (DFDs) analyzing attack vectors across BlazerNET enterprise network and IoT camera endpoints.

> **Impact on Future Career:**
> Instills proactive security engineering instincts to anticipate adversary attack vectors before software deployment. Equips Andrew to evaluate enterprise software against OWASP Top 10, NIST SP 800-53 controls, and federal zero-trust mandates.

University of Montevallo CTE Business Education

Provisional Certificate in a Teaching Field (PCTF) — Business, Marketing & Finance

Provisional Certificate in a Teaching Field: Business, Marketing, and Finance (non-degree), University of Montevallo, Fall 2024 – Spring 2025, GPA 3.66.

> **Impact on Future Career:**
> In federal cybersecurity, analysts must translate complex technical cyber risks to non-technical executives and train personnel on cyber hygiene.

B.S., Biochemistry (ACS), with a secondary major in Biology Medical Sciences; Minor in Music, Mississippi College (now Mississippi Christian University), Aug 2011 – May 2016, cum laude

Medical Studies, Course and clinical work, University of Mississippi School of Medicine, 2016–2020. GPA 3.50.

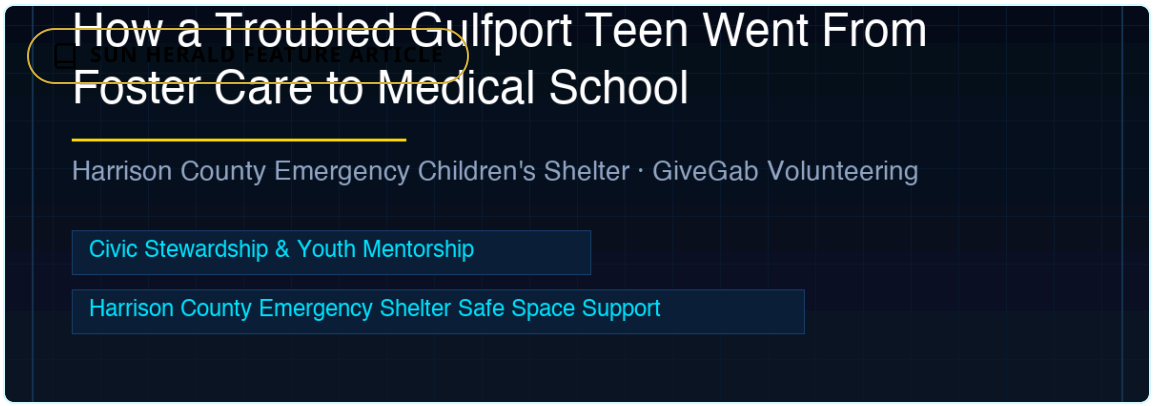
Medical Studies, Course and clinical work. Human Anatomy Prosector (May–Sep 2017). ACS-certified biochemistry laboratory protocols.

> **Impact on Future Career:**
> Instills scientific empirical hypothesis testing and zero-defect attention to detail. Operating in sterile surgical suites and cGMP extraction facilities directly mirrors the discipline required to enforce strict zero-trust postures and safeguard sensitive health data (HIPAA).

Professional Development (2023–2025)

Conferences, technical symposia, specialized certifications, community service leadership, and experiential professional achievements across 2023–2025.

Conferences, Symposia & Community Leadership



COMMUNITY STEWARDSHIP · DECEMBER 2018

Sun Herald Feature: GiveGab Youth Mentorship & Shelter Support

Featured in the Sun Herald for spearheading regional community volunteering and youth advocacy across Harrison County, coordinating mentorship to support vulnerable youth in emergency shelter care.

> **Leadership Impact:**
> Fosters high-trust civic stewardship, digital community engagement, and empathetic leadership foundational to federal public service.

Read Sun Herald Feature Coverage → (https://www.sunherald.com/news/local/counties/harrison-county/article97803862.html)



HEALTHCARE POLICY & CRISIS COUNCIL

Opioid Crisis Council at UMMC & Youth Shelter Advocacy

Collaborated on medical and crisis mitigation initiatives at the University of Mississippi Medical Center (UMMC), addressing community healthcare interventions, harm reduction, and supporting Harrison County's emergency youth shelter as a safe space for children in crisis.

> **Leadership Impact:**
> Bridges clinical systems insight and emergency community care to build trauma-informed, safe environments for at-risk youth.

View WLOX News Safe Space Story → (https://www.wlox.com/2018/12/12/south-mississippi-strong-harrison-countys-emergency-youth-shelter-gives-safe-space-children-need/)

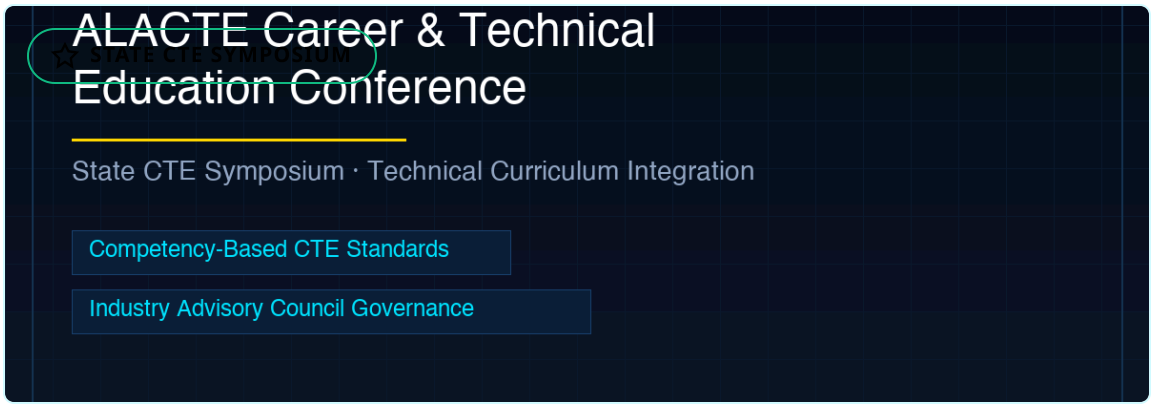


INTERNATIONAL CONFERENCE · ANAHEIM, CA

DECA International Career Development Conference (ICDC)

Completed the *DECA Advisor Professional Learning Series* ("This Is How We Do DECA") at Anaheim Convention Center, Room 207. Mentored rural high school students from new chapter inception to 1st and 2nd place state podium finishes, qualifying for internationals.

> **Professional Growth:**
> Advanced competitive coaching strategies in financial literacy, marketing analytics, entrepreneurship, and executive presentation.

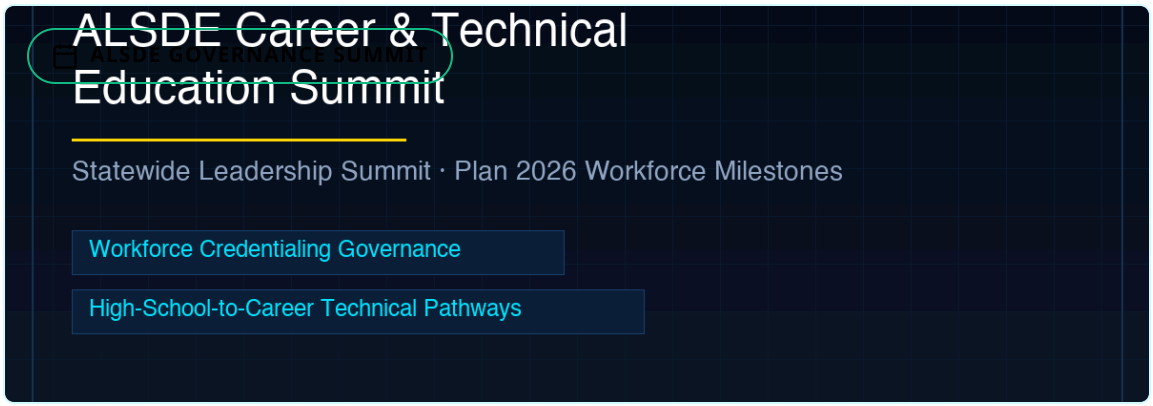


STATE CTE SYMPOSIUM · 2023–2024

ALACTE Career & Technical Education Conference

Participated in the Alabama Association for Career and Technical Education (ALACTE) annual conference, exploring cutting-edge technical curriculum integration, student certification pipelines, and industry advisory council development.

> **Instructional Impact:**
> Implemented competency-based CTE standards aligned with industry demands in business finance and technology.

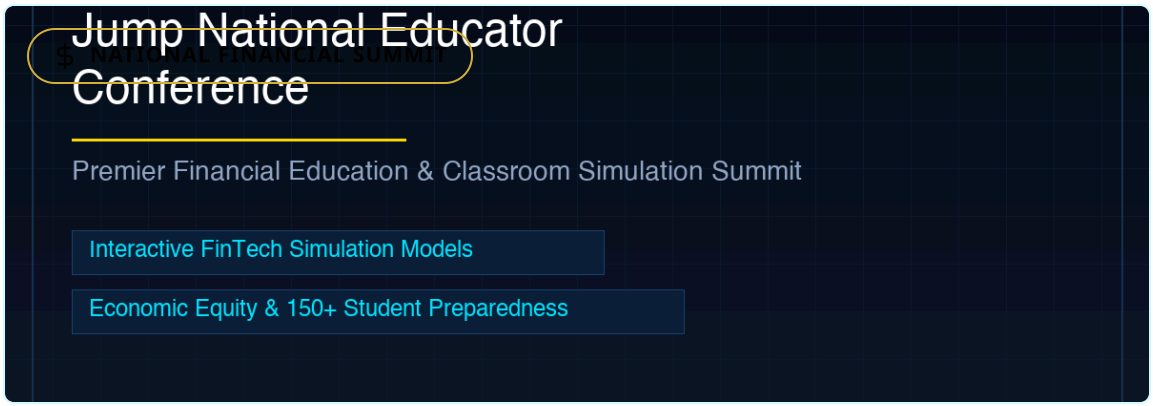


ALSDE GOVERNANCE · 2024–2025

ALSDE Career & Technical Education Summit

Statewide CTE leadership summit focusing on workforce credentialing governance and high-school-to-career technical pathways.

> **Governance Impact:**
> Followed Alabama TAPE evaluation governance and student credential verification.



NATIONAL FINANCIAL SUMMIT · LOUISVILLE, KY

Jump$tart National Educator Conference (Louisville, KY / Kentucky Derby Leadership)

Intensive professional learning on interactive classroom financial simulation tools and FinTech curriculum design.

> **Curricular Innovation:**
> Deployed experiential financial simulation models and cross-sector leadership frameworks that prepared over 150 students for workforce economics.

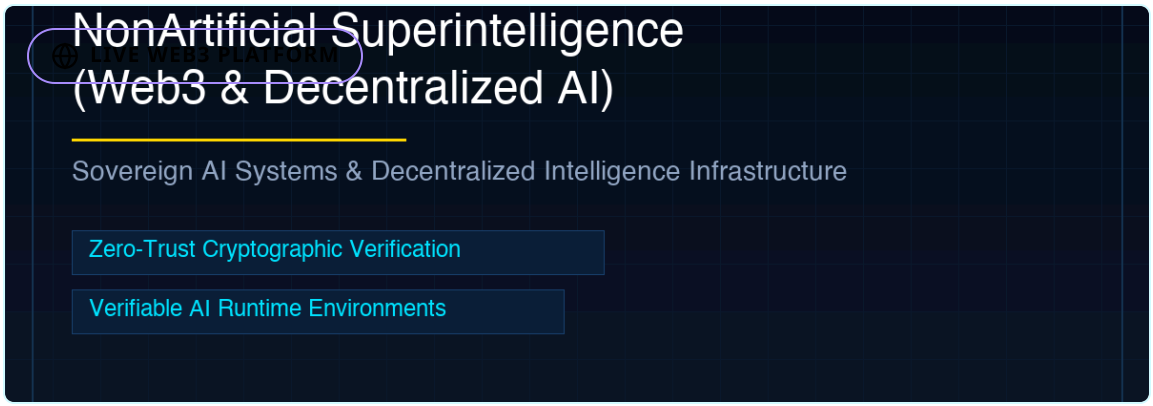


INSTITUTIONAL GOVERNANCE · UMMC

UMMC ASB Quality Improvement Chair for Education

Appointed as Quality Improvement Chair for Education by the UMMC Associated Student Body (ASB) Executive Council. Directed student-faculty academic feedback loops and clinical curriculum quality audits.

> **Institutional Leadership:**
> Governed formal quality improvement methodologies, academic stakeholder communications, and high-stakes administrative reporting in an academic health science center.



FRONTIER AI & WEB3 SYSTEMS · 2024–2025

NonArtificial Superintelligence (Web3 & Decentralized AI)

Architecting decentralized intelligence infrastructure and sovereign AI systems at NonArtificial Superintelligence.

> **Systems Innovation:**
> Bridges zero-trust computational verification with sovereign decentralized architectures, establishing verifiable AI runtime environments.

Explore NonArtificial Superintelligence → (https://nonartificialsi.com)

Federal Career Execution & Public Service Stages

51 Tracked Opportunities

51-Position Federal Opportunity Tracker

Documented in `accurateinternshiptracker.xlsx`: tracks positions across top-choice agencies (White House OSTP, ONCD, USDS), National Labs (Sandia TITANS, Los Alamos CAI, Oak Ridge), and Defense (U.S. Army Space & Missile Defense Command, Redstone Arsenal, CISA).

> **Career Relevance:**

14 Federal Drops

14 SFS Virtual Job Fair Applications

Submitted tailored applications to 14 premier federal intelligence, defense, and civilian agencies: U.S. Space Force, CISA, MARFORCYBER, DOJ CCIPS Cyber Crime Lab, CBP Cyber Threat Intelligence, and Sandia National Laboratories.

> **Career Relevance:**
>
> Direct engagement with federal recruiters securing summer graduate internships that convert into permanent federal civil service roles.

Global Leadership

International Humanitarian Service

Certified Village Health Team (VHT) member, UMMC & OmniMed, Uganda (Jul 2017 – Aug 2017). Mobilized and ran mobile clinics with the UMMC team; helped establish community health teaching groups. U.S. Department of State Benjamin A. Gilman International Scholarship in Pokhara, Nepal conducting hospital clinical service and educational broadcasts for Reach the World.

> **Career Relevance:**
> Cultivates global situational awareness, high-stakes operational composure, and community stewardship foundational to public service leadership.

STEM Mentorship

Youth STEM Drone & Python Mentoring

Python Coding Summer Camp Teaching Assistant, UAB (Jun 2026 – Jul 2026), instructing middle school students in algorithmic coding, BBC micro:bit microcontrollers, and autonomous aerial drone trajectory programming.

> **Career Relevance:**
> Validates hands-on hardware-software integration capabilities and commitment to inspiring the next generation in computer science.

Special Skills & Verified Credentials

Five curated special skills strictly correlated to the federal cybersecurity mission, anchored by an official Credly-verified Microsoft Certified Educator (MCE) digital credential and 30 verified LinkedIn Learning certifications.

Verified Industry Credential

Microsoft Certified Educator (MCE)

Issuer: Microsoft Corporation & Certiport · Issued May 16, 2025 · Verified via Credly

Validates technology literacy and pedagogical competencies needed to architect custom learning environments. Proves Andrew's ability to design enterprise cybersecurity awareness programs, conduct federal workforce upskilling, and translate technical security controls into operational standards.

Verify Credly Badge (Public URL) (https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url)

Top 5 Specialized Skills Aligned to Federal Mission

1. Secure Systems & Zero-Trust Architecture

Credential: SFS CyberAICorps scholarship recipient (NSF-funded, with a post-graduation government service commitment). UAB M.S. Cybersecurity GPA 3.75.

Enforces OMB M-22-09 zero-trust directives through STRIDE threat modeling, packet inspection (Wireshark), port auditing (Nmap), and defense-in-depth principles.

2. Educational Technology & Instructional Design

Credential: Microsoft Certified Educator (Credly Verified) & Alabama PCTF

Architects technical security training programs, translates technical controls into standard operating procedures, and trains non-technical agency personnel.

3. On-Device AI & Privacy-Preserving Systems

Credential: Native Apple Silicon (MaQkrs AI Tutor) & 30 Professional AI Certs

Prevents sensitive data egress by deploying local RAG architectures, quantized Gemma 4 models, and hardware-accelerated inference via Apple MLX.

4. Cloud Infrastructure & DevSecOps Engineering

Credential: Cloud Computing (Summer 2026) and multi-cloud production deployments

Architects least-privilege AWS/GCP/Firebase environments with expiring HMAC token verification, Docker isolation, and automated CI security gates.

5. Regulatory Compliance & Quality Systems

Credential: MidSouth Extracts Operations Director, January 2023 – May 2023

Bridges technical cybersecurity controls with legal and regulatory mandates, applying pharmaceutical cGMP discipline to federal GRC and digital forensics.

Directory of 30 Verified LinkedIn Learning Professional Certifications

All 30 certifications have been verified live (HTTP 200) matching recipient identity and course curricula across Agentic AI, Cloud Core Services, and AI Governance.

- ► Developing Executive Presence (https://www.linkedin.com/learning/certificates/80c5af1302366bc0f58975b8e47366b1b9cacdf4b1717e461be45d11554e637c/)
- ► How to Make Innovative Thinking a Habit (https://www.linkedin.com/learning/certificates/38db059c12bd74c588c52bd46f1620e05e713af07867fc404a8a1b69232ad458/)
- ► Tech Trends (https://www.linkedin.com/learning/certificates/9f02
- ► Fundamentals of Agentic AI (https://www.linkedin.com/learning/certificates/a54

c8b5d449d36f700357c322b937c3046baebaa03f45daa07441f7a09cf018/)

- Generative AI with Multi-Agent LangChain (https://www.linkedin.com/learning/certificates/4de319facbd958b86695aed10dd1d63daaf831340a1d69606402ab82303e37f5/)
- How to Boost Productivity with AI (https://www.linkedin.com/learning/certificates/44d54d96d179aea9bf479197251238c5042b686ebe2594cca702eff0a408a346/)
- Nano Tips for Using AI (Rachel Woods) (https://www.linkedin.com/learning/certificates/11d0967544ec3840b7f7bec263760ecdb363bf12ac868a24e34c47f023fa6c47/)
- What Is Generative AI? (https://www.linkedin.com/learning/certificates/cb373b709a8b3183a20256864583e114b05f29331dd4801004799b7212373799/)
- Generative AI in Finance & Accounting (https://www.linkedin.com/learning/certificates/62c8b62d8af996b7c79229cbc1c24818b26717e99598b567efe66ae5476dab84/)
- Accounting Foundations: Bookkeeping (https://www.linkedin.com/learning/certificates/6e9c8da478167e6b25f97982302053aac42c2bb79307726af7a14f0ca0aa8624/)
- Agentic AI Business Strategy (Charlene Li) (https://www.linkedin.com/learning/certificates/ee7794b7f7c55f34897761d1dc61991c6e17bf1ce18af0ae9831b2a81d4aceee/)

0ca10a00b2bb365732a59e9a3586dfbc44f3a5c6682a591c9a9cf1faba6ad/)

- Building Agents with Vertex AI (https://www.linkedin.com/learning/certificates/f64c67bcb95f256f4628665c9b3e0b1716fd10c388f540e32338a6c00bb5acfa/)
- Greg McKeown on Effortless Results (https://www.linkedin.com/learning/certificates/31f95bc84f20097cfa91e5161db36389ef7d09fca0060036d4d172b8e3269e95/)
- Power Platform Fundamentals (PL-900) (https://www.linkedin.com/learning/certificates/cf971c20e18b737b4e4192091dc931a7247b81b99bab07a264e442ae2c59e3d4/)
- Generative AI vs. Traditional AI (https://www.linkedin.com/learning/certificates/dddfc1f7c9dba1efe07f16d7b3675b7c6de62a644eb0fdc38199a710212d4dcb/)
- Accounting Foundations (https://www.linkedin.com/learning/certificates/486e8a57a5d1394cdff715d976fe07abb14c3fa5a2a8159453d71cd9855fa615/)
- Advanced AI Governance & Controls (https://www.linkedin.com/learning/certificates/91b82fcde090789768b336db8fce7914e110627c9a3fa64c77a962f91f9bfcfd/)
- Copilot for Service (https://www.linkedin.com/learning/certificates/e114997ddcd93805e977dda58fd043f85767e23706cec682ddc58e20dd064e70/)

- ▶ AI for React Developers (https://www.linkedin.com/learning/certificates/397b40ed680f95dc12e184dc64aeda24e1c09ac66a3e6de9db32e685dcc16990/)
- ▶ Knowledge Graphs for Generative AI (https://www.linkedin.com/learning/certificates/212aa3ec812bc240416c88a219255fe3a0dd5d439b0dc9d1d81669617cce86d5/)
- ▶ Google Cloud Core Services (https://www.linkedin.com/learning/certificates/a91915bc420b3f58d1fc44e3398b1e6f026afc87d4cb85975d4f8cda82fa7590/)
- ▶ Agentic AI for Developers (https://www.linkedin.com/learning/certificates/32befa0c9bbe8cda3cbe4d7afdd748ca6cb51bc0ed9fd8d857182f67e35a1baa/)
- ▶ Preparing Your Organization for AI Agents (https://www.linkedin.com/learning/certificates/c858777de5b940f5a9bb20b6d3a81f2c0052948978593e13156a961f3c528be2/)
- ▶ Upskilling Your Organization on AI (https://www.linkedin.com/learning/certificates/ebdbdcec76b0c8ea565624a89a4db6c30c25cfb520f45aac9c74dd8d08f207a0/)
- ▶ Future of Workforce Learning (https://www.linkedin.com/learning/certificates/e727c26a36b402856aff5af64ba73fb8ae2cd74f56cad42358c15c1be1a1c004/)
- ▶ Generative AI in Learning & Development (https://www.linkedin.com/learning/certificates/d3fc3390e22c5d64da8d7cec42a72e054d8346cecdeb603aaa8feec7a332e3fd/)
- ▶ Using Gen AI for Personalized Learning (https://www.linkedin.com/learning/certificates/b0b910f867ca5b4a18da366bfc120d2d8dcc10fd2aed077a74e7f4d222a70e2f/)
- ▶ Building AI Apps with Prompt Flow (https://www.linkedin.com/learning/certificates/638879edcc13d8dd4d774904da536a25da1418e0a973d4ddeca651d477e6ba10/)
- ▶ Finding & Retaining High Potentials (https://www.linkedin.com/learning/certificates/ef48d5d0458c6d9f06c3a45f6980c588925880b2fc881e7aba55367e8bc27344/)
- ▶ Google Gemini for Business Analysis (https://www.linkedin.com/learning/certificates/bab98e0f5779c30857cbd9ce5313d778d772e467596604de6a1c15ce2fba7303/)

Interactive Media & 28+ Project Showcase

Explore Andrew's real technical body of work. Inspect embedded 10-second web video previews with dual MP4/WebM formats, followed by a verified live directory of 28+ cloud-hosted applications.

Curated Highlight Projects with Video Previews

CS646 Sanctum 3D Memory Palace

Gamified Cryptographic Distributed Ledger Environment

Godot 4 | WebGL | AWS CloudFront | SHA-256 | TypeScript

Interactive 3D exploration world featuring 5 distinct districts, procedural hover traversal, and real-time SHA-256 block hashing demonstrations.

MaQkrs AI Tutor (Local AI Study Assistant)

Native Apple Silicon Local RAG Study Assistant

SwiftUI | SwiftData | Apple MLX | Quantized Gemma 4 | Ollama

Native assistant indexing documents and code for semantic query retrieval with 0 network egress, designed to keep data on the device.

AdaptiveSchool

A schooling portal for students, teachers, and parents

Next.js 15 | Firebase Firestore | Cloud Run | Tailwind CSS | DemoAuth

A schooling portal for students, teachers, and parents.

MaQkrs OS Command Center & Planner

Executive Dashboard & Multi-Agent Operations Console

React 19 | TypeScript | Firebase | Tailwind CSS | Python

Unified operational command center providing real-time telemetry across multi-agent workflows, business financials, and project sprint schedules.

Directory of 28+ Verified Live Project Deployments

Every project link below is live-checked (HTTP 200) across AWS CloudFront, Firebase Hosting, Vercel, and Lovable platforms.

CS646 Sanctum 3D World (AWS)

(https://d1puxvoeqwchkj.cloudfront.net/)

AWS CloudFront · Godot WebGL

Primary 3D memory palace exploring cryptographic consensus.

CS646 Sanctum (Firebase Fallback)

(https://cs646-sanctum-world.firebaseapp.com/)

Firebase Hosting · WebGL

Alternative high-availability node for the Sanctum 3D world.

CS646 Blockchain Study Hub

(https://cs646-sanctum-astrachan.firebaseapp.com/)

Firebase Hosting · React

Permanent course portal for UAB CS 646 Blockchain & Cryptocurrency.

CS646 Midterm Review Portal

(https://cs646-sanctum-astrachan--midterm-aogm2izm.web.app/#/)

Firebase Web App · Single Page

Cryptographic lexicons, study guides, and flashcards.

Quantum Universe 3D Dashboard
(https://quantum-universe-3d.web.app/)
Firebase Web App · Three.js

Cyber-themed spatial operations console with live nodes.

MaQkrs OS Command Center
(https://maqkrs-command-center-i.firebaseapp.com/)
Firebase Hosting · React 19

Multi-agent business operations and schedule dashboard.

AdaptiveSchool **(https://adaptivehs-v2.firebaseapp.com/)**
Cloud Run · Next.js 15

A schooling portal for students, teachers, and parents.

Project Atlas 3D Portfolio
(https://d3jeotfnsm148g.cloudfront.net/)
AWS CloudFront · Spatial Storytelling

Scroll-driven canvas architecture with interactive case studies.

Coastal Speech Co. **(https://coastal-speech-co-fl.firebaseapp.com/)**
Firebase Hosting · React

Clinical practice web platform for Panama City, FL speech clinic.

Victorious Herbal Elements
(https://herbal.nonartificialsi.com/)
Vercel · Next.js Storefront

Client botanical product catalog and checkout workflow.

Networking Universe (CS 334/534)
(https://networking-testy.firebaseapp.com/)
Firebase Hosting · 3D Visualizer

Interactive 3D network topology and packet routing simulation.

CS 623 Network Security Hub
(https://networking-security-cs623-hub.firebaseapp.com/)
Firebase Hosting · Markdown Hub

Course resource platform for firewalls and network defense.

Switch MAC Address Learning
(https://practice-questions-networking.firebaseapp.com/)
Firebase Hosting · Interactive Canvas

Interactive Layer 2 switch forwarding table simulator.

Networking Midterm Guide
(https://networking-midterm.web.app/studyguide4.html)
Firebase Web App · Deep Link

Intensive study guide with 33 question variant modules.

CS 334/534 Study Resources
(https://networking-exam-60418.web.app/)
Firebase Web App · Static Hub

Static reference platform organizing network exam guides.

CS 423/623/723 Practice Test
(https://network-security-study.firebaseapp.com/)
Firebase Hosting · 200-Q Test

Network security examination engine testing defense-in-depth.

Network Security Demos (https://ns-demos-34811.firebaseapp.com/)
Firebase Hosting · State Engine

Interactive firewall, proxy, and access control list (ACL) demos.

Cyber Risk Management (NIST)
(https://cyberriskmanagement-testy.firebaseapp.com/)
Firebase Hosting · NIST Portal

UAB risk study hub mastering NIST SP 800-30 and CSF.

CS 203 OOP Study Hub (https://cs203-java-studyguide.firebaseapp.com/)
Firebase Hosting · Code Snippets

Java OOP guide covering inheritance and encapsulation.

CS 203 Practice Question Bank
(https://cs203-java-practice1.firebaseapp.com/)
Firebase Hosting · Question Bank

Interactive Java OOP drill examining object reference models.

CS 203 OOP Exam Simulator
(https://cs203-exam-simulator.firebaseapp.com/)
Firebase Web App · Timed Simulator

Timed exam simulator grading complex Java programs.

CS 203 Code Output Analysis (https://cs-203-midterm.web.app/practice_test_labs.html)
Firebase Web App · Deep Path

Diagnostic tool testing student ability to trace Java code.

CS 203 Exam 1 Practice Test (https://cs-203-midterm.web.app/practice_test.html)
Firebase Web App · Practice Test

Practice test covering loops, class inheritance, and types.

Computer Security Companion (https://secure-connector-buddy.lovable.app/)
Lovable App · AI Study Tool

Interactive AI companion for CS 436/636/736 computer security.

Cybersecurity Study Guide (https://andy-sg-ns.firebaseapp.com/)
Firebase Hosting · Question Engine

Question variant generator covering security fundamentals.

GHS Learning Platform (https://ghs-lp-dev-astrachan163.firebaseapp.com/)
Firebase · Google Cloud Prototype

Functional testing prototype for student portal and course delivery.

Sources & Provenance Ledger

Full audit trail substantiating every factual claim, credential, labor statistic, and code artifact against professionally legitimate authorities (BLS, NIST, NSF, Microsoft Credly).

Authoritative Citations & Federal References

- **U.S. Bureau of Labor Statistics (BLS):** *Occupational Outlook Handbook: Information Security Analysts.* Data release 2024–2026. https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm (https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm) (SOC 15-1212).
- **National Center for O*NET Development:** *15-1212.00 – Information Security Analysts.* O*NET OnLine. Sponsored by USDOL/ETA. https://www.onetonline.org/link/summary/15-1212.00 (https://www.onetonline.org/link/summary/15-1212.00).
- **National Science Foundation (NSF):** *CyberCorps: Scholarship for Service (SFS) Program Guidelines.* Directorate for STEM Education. https://new.nsf.gov/funding/opportunities/cybercorps-scholarship-service-sfs (https://new.nsf.gov/funding/opportunities/cybercorps-scholarship-service-sfs) and https://www.sfs.opm.gov/ (https://www.sfs.opm.gov/).
- **National Institute of Standards and Technology (NIST):** *Security and Privacy Controls for Information Systems and Organizations.* NIST SP 800-53, Rev. 5.

https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final (https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final).

- **Microsoft Corporation & Certiport:** *Microsoft Certified Educator (MCE) Digital Badge Verification.* Credly Digital Credential Platform. https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url (https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url).
- **Harrison County Youth Outreach & Giving Community:** *Harrison County Community Support Initiatives.* https://www.sunherald.com/news/local/counties/harrison-county/article97803862.html (https://www.sunherald.com/news/local/counties/harrison-county/article97803862.html).
- **South Mississippi Emergency Youth Shelter & Healthcare Outreach:** *Harrison County Emergency Youth Shelter & UMMC Council.* https://www.wlox.com/2018/12/12/south-mississippi-strong-harrison-countys-emergency-youth-shelter-gives-safe-space-children-need/ (https://www.wlox.com/2018/12/12/south-mississippi-strong-harrison-countys-emergency-youth-shelter-gives-safe-space-children-need/).

Andrew Strachan

ELECTRONIC CAREER PORTFOLIO

Personal Electronic Career Portfolio for an M.S. Cybersecurity student and builder.

Contact: astrachan162@gmail.com · Birmingham, Alabama

PORTFOLIO SECTIONS

Interactive Resume

Career Research (SOC 15-1212.00)

Career-Related Education

Professional Development

Special Skills & MCE Credential

28+ Projects & Media Showcase

Sources & Provenance Ledger

AUTHORITATIVE VERIFICATIONS

Credly MCE Badge Verification (https://www.credly.com/badges/d4e5c326-c255-405c-b50e-0a369d6fc3a0/public_url)

NSF CyberCorps SFS Program (https://new.nsf.gov/funding/opportunities/cybercorps-scholarship-service-sfs)

BLS Occupational Outlook Handbook (https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm)

GitHub: @astrachan163 (https://github.com/astrachan163)

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